

Two-dimensional heat (

Two-dimensional heat equation
(Laplace eqn) under steady state condition

One-dimensional heat eqn

$$\frac{\partial y}{\partial t} = c^2 \frac{\partial^2 y}{\partial x^2}$$

Now two-dim heat eqn

$$\frac{\partial y}{\partial t} = c^2 \left(\frac{\partial^2 y}{\partial x^2} + \frac{\partial^2 y}{\partial y^2} \right)$$

Steady - state condition means the
state at which the temp = constant.

In the Steady state condition, we have

$$\frac{\partial y}{\partial t} = 0 \text{ so the eqn reduces}$$

$$\boxed{\frac{\partial^2 y}{\partial x^2} + \frac{\partial^2 y}{\partial y^2} = 0}$$

which is called the Laplace equation in two-dimension.

Solution of Laplace Equation:-

Let $u(x, y) = X(x) \cdot Y(y)$ let the

$$\text{soln of } \frac{\partial^2 y}{\partial x^2} + \frac{\partial^2 y}{\partial y^2} = 0 \text{ --- (1)}$$

$$\Rightarrow Y X'' + X Y'' = 0$$

$$\Rightarrow Y X'' = -X Y''$$

$$\Rightarrow \frac{X''}{X} = -\frac{Y''}{Y} = k \text{ (say)}$$

$$\Rightarrow \left. \begin{aligned} X'' - kX &= 0 \\ Y'' + kY &= 0 \end{aligned} \right\} \text{ --- (2)}$$

Case (1)

when $k = 0$ then (2) becomes

$$X'' = 0$$

$$Y'' = 0$$

$$\Rightarrow X = 4x + c_2 \quad | \quad Y = c_3 y + c_4$$

$$\therefore u = XY = (4x + c_2)(c_3 y + c_4) \quad \text{--- (3)}$$

Case (2) when $k > 0$ i.e. $k = b^2$

from (2)

$$X'' - b^2 X = 0$$

$$\Rightarrow (D^2 - b^2) X = 0$$

$$\Rightarrow m^2 - b^2 = 0$$

$$\Rightarrow m = \pm b$$

$$X = c_1 e^{bx} + c_2 e^{-bx}$$

$$Y'' + b^2 Y = 0$$

$$(D^2 + b^2) Y = 0$$

$$\Rightarrow m_1^2 + b^2 = 0$$

$$\Rightarrow m_1 = \pm bi$$

$$Y = c_1 \cos by + c_2 \sin by$$

$$\therefore u = XY$$

$$\Rightarrow u = (c_1 e^{bx} + c_2 e^{-bx})(c_1 \cos by + c_2 \sin by)$$

--- (4)

Case (3)

when k is -ve i.e. $k = -b^2$ then (2)

becomes

$$\Rightarrow X'' + b^2 X = 0$$

$$\Rightarrow D^2 + b^2 = 0$$

$$\Rightarrow D = \pm bi$$

$$X = (c_1 \cos bx + c_2 \sin bx)$$

$$Y'' - b^2 Y = 0$$

$$D^2 - b^2 = 0$$

$$D = \pm b$$

$$Y = c_1 e^{by} + c_2 e^{-by}$$

$$\Rightarrow u(x, y) = xy$$

$$\Rightarrow u = (c_1 \cos bx + c_2 \sin bx) (c_3 e^{by} + c_4 e^{-by}) \quad \text{--- (5)}$$

Hence for the Laplace eqn

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \quad \text{we get -}$$

$$1) u(x, y) = (c_1 x + c_2) (c_3 y + c_4)$$

$$2) u(x, y) = (c_1 e^{bx} + c_2 e^{-bx}) (c_3 \cos by + c_4 \sin by)$$

$$3) u(x, y) = (c_1 \cos bx + c_2 \sin bx) (c_3 e^{by} + c_4 e^{-by})$$

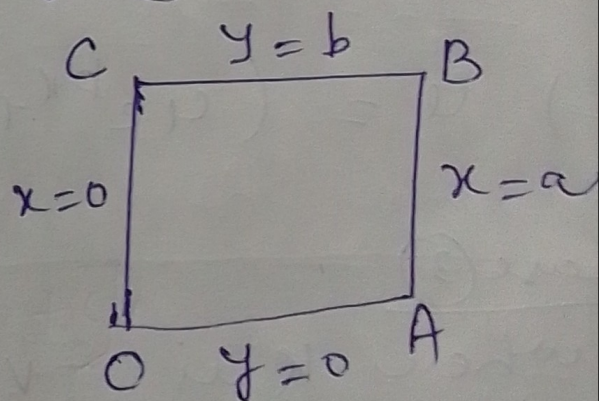
Boundary conditions are

$$u(0, y) = u_1$$

$$u(a, y) = u_2$$

$$u(x, 0) = u_3$$

$$u(x, b) = u_4$$



Q Solve the Laplace equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \text{ subject to the conditions}$$
$$u(0, y) = 0, \quad u(l, y) = 0,$$
$$u(x, 0) = 0 \quad \text{and} \quad u(x, a) = \sin \frac{n\pi x}{l}.$$

Sol Given $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ ——— (1)

Let the soln of (1)

$$u(x, y) = (c_1 \cos bx + c_2 \sin bx)(c_3 e^{by} + c_4 e^{-by})$$

————— (2)

conditions are

$$u(0, y) = 0$$

$$u(l, y) = 0$$

$$u(x, 0) = 0$$

$$u(x, a) = \sin \frac{n\pi x}{l}$$

Put $x=0$ in (2)

$$\Rightarrow 0 = c_1 (c_3 e^{by} + c_4 e^{-by})$$

$$c_3 e^{by} + c_4 e^{-by} \neq 0, \quad c_1 = 0.$$

then eqn (2)

$$\Rightarrow u = c_2 \sin bx (c_3 e^{by} + c_4 e^{-by})$$

————— (3)

$$u(l, y) = 0$$

$$\Rightarrow \text{Put } x = l \text{ in (3)}$$

$$0 = c_2 \sin bl [c_3 e^{by} + c_4 e^{-by}]$$

$\Rightarrow$ if $c_2 = 0$ then soln does not exist.

$$\text{so } \sin bl = 0 \Rightarrow \sin bl = \sin n\pi$$

$$bl = n\pi \Rightarrow b = \frac{n\pi}{l}$$

using in (3)

$$\Rightarrow u = c_2 \sin \frac{n\pi}{l} x \left[c_3 e^{\frac{n\pi}{l} y} + c_4 e^{-\frac{n\pi}{l} y} \right] \quad \text{--- (4)}$$

$$\text{Now } u(x, 0) = 0$$

$$\text{Put } y = 0 \text{ in (4)}$$

$$0 = c_2 \sin \frac{n\pi x}{l} [c_3 + c_4]$$

$$\Rightarrow c_3 + c_4 = 0, \quad c_2 \neq 0$$

$$\Rightarrow c_4 = -c_3$$

using in (4)

$$u(x, y) = c_2 c_3 \sin \frac{n\pi x}{l} \left[e^{\frac{n\pi y}{l}} - e^{-\frac{n\pi y}{l}} \right] \quad \text{--- (5)}$$

Most general form is

$$u(x, y) = b_n \sin \frac{n\pi x}{l} \left[e^{\frac{n\pi y}{l}} - e^{-\frac{n\pi y}{l}} \right] \quad \text{--- (5)}$$

$$u(x, a) = \sin \frac{n\pi x}{l}$$

Put $y = a$ in (5)

$$\cancel{\sin \frac{n\pi x}{l}} = b_n \cancel{\sin \frac{n\pi x}{l}} \left[e^{\frac{n\pi y}{l}} - e^{-\frac{n\pi y}{l}} \right]$$

$$\Rightarrow b_n = \frac{1}{e^{\frac{n\pi y}{l}} - e^{-\frac{n\pi y}{l}}}$$

using in (5)

$$\Rightarrow u(x, y) = \frac{1}{e^{\frac{n\pi y}{l}} - e^{-\frac{n\pi y}{l}}} \sin \frac{n\pi x}{l} \left[e^{\frac{n\pi y}{l}} - e^{-\frac{n\pi y}{l}} \right]$$

$$\text{after putting } a, b_n = \frac{1}{e^{\frac{n\pi a}{l}} - e^{-\frac{n\pi a}{l}}}$$

So

$$u = \frac{1}{e^{\frac{n\pi a}{l}} - e^{-\frac{n\pi a}{l}}} \sin \frac{n\pi x}{l} \left[e^{\frac{n\pi y}{l}} - e^{-\frac{n\pi y}{l}} \right] \quad \text{Ans.}$$